

MOCK EXAMINATION — TRY BEFORE YOU BOOK

GCSE Physics

Mock Test & Mark Scheme

Time allowed	25 minutes
Total marks	38 marks
Number of questions	27
Topics covered	Forces & Motion, Energy, Electricity, Waves

Student name: _____ **Date:** _____

Instructions

- Answer every question in the space provided on your own paper.
- Write your name and today's date at the top of your answer sheet.
- The marks for each question are shown in brackets, e.g. **(2 marks)**.
- Where a calculation is required, show your working — marks are available for method as well as the final answer.
- Once you finish, check your answers against the mark scheme at the back of this document.
- This mock test is provided free by BrainyEdHub to give you a feel for our teaching standard — book a free trial class to go through it with a tutor.

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Question Paper

Forces & Motion

1. State the unit of force. [1 mark]
2. Write the equation linking force, mass and acceleration. [1 mark]
3. A 5 kg object accelerates at 2 m/s^2 . Calculate the force acting on it. [2 marks]
4. A car travels 150 m in 10 s. Calculate its average speed. [2 marks]
5. Write the equation for calculating momentum. [1 mark]
6. State Newton's First Law of Motion. [2 marks]
7. Calculate the weight of a 10 kg object ($g = 10 \text{ N/kg}$). [2 marks]

Energy

8. Define 'kinetic energy' and give its formula. [2 marks]
9. Calculate the kinetic energy of a 2 kg object moving at 3 m/s. [2 marks]
10. State the principle of conservation of energy. [2 marks]
11. Name the energy store gained by an object when it is raised above the ground. [1 mark]
12. Calculate the gravitational potential energy of a 2 kg object raised 5 m ($g = 10 \text{ N/kg}$). [2 marks]
13. Name one way energy can be usefully transferred electrically. [1 mark]

Electricity

14. State the unit of electrical current. [1 mark]
15. Write the equation linking voltage, current and resistance. [1 mark]
16. A circuit has a current of 2 A and a resistance of 5Ω . Calculate the voltage. [2 marks]
17. State one difference between a series circuit and a parallel circuit. [2 marks]
18. Name the component used to measure current in a circuit. [1 mark]
19. Name the component used to measure potential difference in a circuit. [1 mark]
20. State what happens to the resistance of a wire as its length increases. [1 mark]

Waves

21. State whether sound waves are longitudinal or transverse. **[1 mark]**
22. Define 'wavelength'. **[1 mark]**
23. Calculate the wave speed of a wave with frequency 50 Hz and wavelength 4 m. **[2 marks]**
24. State whether light waves are longitudinal or transverse. **[1 mark]**
25. Define the 'frequency' of a wave. **[1 mark]**
26. State the unit of frequency. **[1 mark]**
27. Name the part of the electromagnetic spectrum with the longest wavelength. **[1 mark]**

Mark Scheme

Use this guide to check your answers. Award full marks for correct final answers; where working is shown, award partial credit as indicated.

Q	Model answer / mark scheme	Marks
1	Newton (N)	1
2	$F = m \times a$	1
3	10 N $F = 5 \times 2.$	2
4	15 m/s speed = distance $\div$ time = $150 \div 10.$	2
5	momentum = mass $\times$ velocity ($p = mv$)	1
6	An object stays at rest, or moving at a constant velocity, unless acted on by a resultant force.	2
7	100 N weight = mass $\times$ gravitational field strength = $10 \times 10.$	2
8	The energy an object has due to its motion; $KE = \frac{1}{2}mv^2.$	2
9	9 J $KE = 0.5 \times 2 \times 3^2 = 9 \text{ J}.$	2
10	Energy cannot be created or destroyed, only transferred from one store to another.	2
11	Gravitational potential energy store	1
12	100 J $GPE = \text{mass} \times g \times \text{height} = 2 \times 10 \times 5.$	2
13	E.g. through a circuit to a motor, heater, or lamp.	1
14	Ampere (A)	1
15	$V = I \times R$	1
16	10 V $V = 2 \times 5.$	2
17	In series there's a single path so the current is the same everywhere; in parallel there are multiple paths and the current splits between branches.	2
18	An ammeter	1
19	A voltmeter	1
20	The resistance increases.	1

Q	Model answer / mark scheme	Marks
21	Longitudinal	1
22	The distance from one point on a wave to the identical point on the next wave (e.g. peak to peak).	1
23	200 m/s wave speed = frequency $\times$ wavelength = 50×4 .	2
24	Transverse	1
25	The number of waves passing a point per second.	1
26	Hertz (Hz)	1
27	Radio waves	1

Total for Physics
38 marks

How did you do? A BrainyEdHub tutor can go through any question you got stuck on, and build a personalised revision plan around your results. Book a free trial class at brainyedhub.com/contact.html or message us on WhatsApp.